

Overview

It was once thought that atoms are indivisible. We now know that atoms are made of protons, neutrons and electrons. We also know that protons and neutrons are made of **quarks**. We currently believe that quarks are not made of anything else – they are '**fundamental**' particles. Electrons are also fundamental particles. There are other fundamental particles which are not found in atoms; these include the photon, muon, and higgs particle. There are other particles, made up from quarks. Most can only exist for fractions of a second and are sometimes called 'exotic' particles. **The standard model contains all the fundamental particles.**

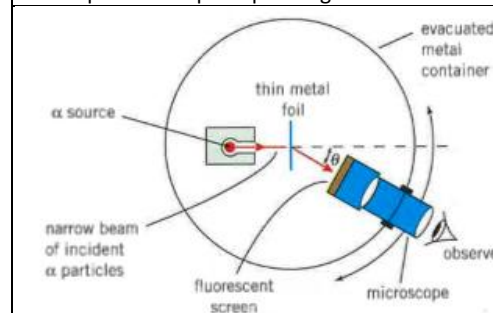
Standard model	The set of all known fundamental particles is called the standard model. This model has stood for several decades now, (with some additions).
Fundamental particle	A particle not made up of anything else
Antiparticle	Every fundamental particle has an antiparticle, with the same mass but opposite charge
Hadron	A particle which experiences the strong nuclear force. Made of quarks
Lepton	A particle which does not experience the strong nuclear force. Leptons are fundamental particles
Baryon	Sub-set of hadrons; made of 3 quarks
Meson	Sub-set of hadrons; made of a quark and an antiquark

'Baryon', 'meson' and 'lepton' mean 'heavy', 'medium' and 'light', from greek. While this is generally true, some leptons are heavier than some mesons so the names are misleading.

Quark	A fundamental particle; they join together to make hadrons
Plum pudding model	In the 19 th century before the discovery of the nucleus, it was thought that atoms were fairly uniform in density, with electrons spread inside them – like plums in a sponge pudding
isotope	Nuclei* of the same element with different numbers of neutrons – eg ^3_2He and ^4_2He are two isotopes of helium. * 'atoms' would also be correct, but we tend to ignore the electrons in their shells in nuclear physics
Fundamental forces	There are only 4 types of force in the universe.
Strong nuclear	The force holding protons and neutrons together in the nucleus. Only affects hadrons. Strongest, but shortest range
Weak nuclear	Causes beta decay. Often called weak nuclear 'interaction'
Electro-magnetic	Electrostatic and magnetic forces, experienced by charged particles.
Gravitational	Affects all particles with mass. Weakest, but longest range.

1. Alpha particle scattering experiment

Rutherford's experiment gave strong evidence that atoms have a nucleus, and are otherwise almost entirely empty. Since all matter is made of atoms it means that all matter is almost entirely empty space – a very surprising finding. This disproves the plum pudding model.



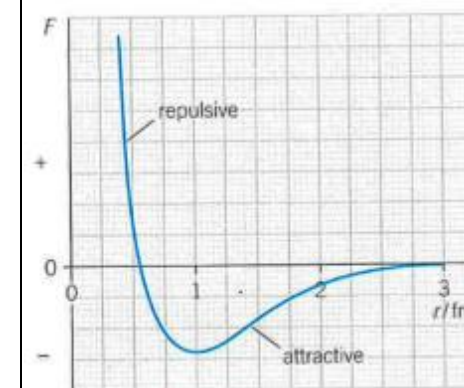
▲ Figure 1 Apparatus for the alpha-scattering experiment of Rutherford, Geiger, and Marsden

Method:	Alpha particles (He nucleus; $2p2n$) were fired at thin gold foil
Findings	Almost all went straight through. Out of every 2000 particles, 1 was deflected. One out of 10000 were deflected by more than 90°
Conclusions	There is a very small, positive charge with a large mass in each atom – now called the nucleus The rest of the atom is empty (apart from the electrons)

If the nucleus were the size of a football, the atom would be the size of a large stadium - completely empty apart from the orbiting electrons – which are all much smaller than the nucleus.

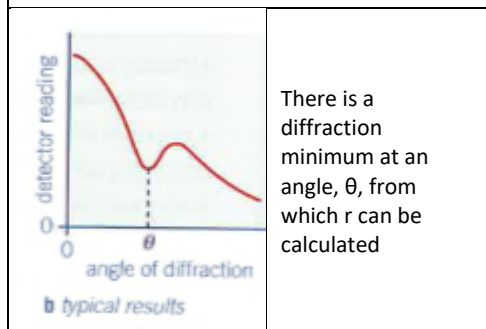
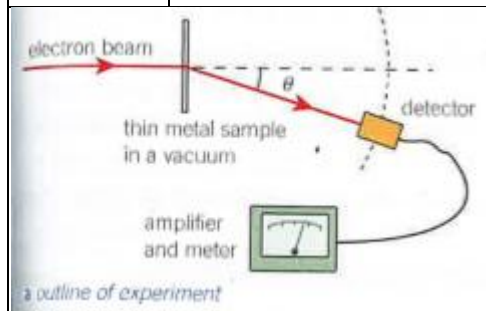
2. The nucleus

Isotope notation	A_ZX X is the element symbol Z is the proton number / atomic number A is the mass number (neutron number, $N = A - Z$)
Atomic mass unit	A non-SI unit often used in nuclear physics; defined as 1/12 of the mass of a C-12 atom. Roughly the mass of a proton or neutron.
Strong nuclear force	Holds the p and n together. It has a strong attraction at a range of 1fm



▲ Figure 3 A graph showing how the nuclear force F varies with separation r for two nucleons

Nuclear size and density	Electron diffraction (see 13.4) is used to find the radii of nuclei. It is found that the radius of a nucleus of mass number A is $R = r_0 A^{1/3}$ Where $r_0 = 1.2 \times 10^{-15} \text{m}$
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3. Antiparticles, hadrons, and leptons	
Antimatter	Matter made of antiparticles.
Positron	The first antiparticle discovered, the antiparticle of an electron. No other antiparticle has its own name, they are just called antiproton, antiquark, etc

4. Quarks	
Protons are made of three quarks (uud) as are neutrons (udd). All other hadrons (both mesons and baryons) are also made of quarks	

5. Beta decay	
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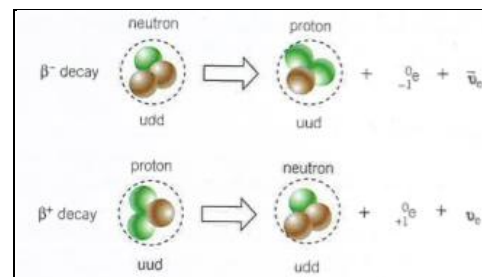
Beta- decay	Occurs naturally when a nucleus is neutron-rich (too many protons for stability) A neutron in the nucleus decays into a proton, releasing an electron and an anti-electron neutrino: ${}_0^1n \rightarrow {}_1^1p + {}_{-1}^0e + \bar{\nu}_e$ In terms of quarks: $udd \rightarrow uud + {}_{-1}^0e + \bar{\nu}_e$ Or more simply: $d \rightarrow u + {}_{-1}^0e + \bar{\nu}_e$
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Beta+ decay	Occurs naturally when a nucleus is proton-rich (too many protons to be stable). A proton in the nucleus decays into a neutron, releasing a positron and an electron neutrino: ${}_1^1p \rightarrow {}_0^1n + {}_{+1}^0e + \nu_e$ In terms of quarks: $uud \rightarrow udd + {}_{+1}^0e + \nu_e$ Or more simply: $u \rightarrow d + {}_{+1}^0e + \nu_e$
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Note: '+1', not just '1', should be used for the positron, for clarity

The electrons come from the decay inside the nucleus – they are **not** from the orbitals/shells!

Beta+ particle is a positron emitted from a nucleus; beta-particle is an electron emitted from a nucleus.



There are conservation rules which these decays obey: charge and nucleon number are conserved. There are other conservation rules (eg lepton number is conserved) which help to understand these decays, these are not included in the course.

Weak nuclear force	The weak nuclear force is responsible for these decays. There is no simple way to visualise or describe it as a force, so it is sometimes called the weak nuclear interaction.
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The **standard model** is the set of all known fundamental particles. Not all are mentioned on this A level course. For the course, the two important groups are first column of quarks and leptons.

Standard Model of Elementary Particles									
three generations of matter (fermions)						interactions / force carriers (bosons)			
I		II		III		interactions / force carriers (bosons)			
mass charge spin									
$\approx 2.2 \text{ MeV}/c^2$ $\frac{2}{3}$ $\frac{1}{2}$	u up	$\approx 1.28 \text{ GeV}/c^2$ $\frac{2}{3}$ $\frac{1}{2}$	c charm	$\approx 173.1 \text{ GeV}/c^2$ $\frac{2}{3}$ $\frac{1}{2}$	t top	0 0 1	g gluon	$\approx 124.97 \text{ GeV}/c^2$ 0 0 0	H higgs
$\approx 4.7 \text{ MeV}/c^2$ $-\frac{1}{3}$ $\frac{1}{2}$	d down	$\approx 96 \text{ MeV}/c^2$ $-\frac{1}{3}$ $\frac{1}{2}$	s strange	$\approx 4.18 \text{ GeV}/c^2$ $-\frac{1}{3}$ $\frac{1}{2}$	b bottom	0 0 1	gamma photon		
$\approx 0.511 \text{ MeV}/c^2$ -1 $\frac{1}{2}$	e electron	$\approx 105.66 \text{ MeV}/c^2$ -1 $\frac{1}{2}$	mu muon	$\approx 1.7768 \text{ GeV}/c^2$ -1 $\frac{1}{2}$	tau tau	0 1 1	Z Z boson	$\approx 91.19 \text{ GeV}/c^2$	
$< 1.0 \text{ eV}/c^2$ 0 $\frac{1}{2}$	nu_e electron neutrino	$< 0.17 \text{ MeV}/c^2$ 0 $\frac{1}{2}$	nu_mu muon neutrino	$< 18.2 \text{ MeV}/c^2$ 0 $\frac{1}{2}$	nu_tau tau neutrino	$\approx 80.39 \text{ GeV}/c^2$ ± 1 1	W W boson		
LEPTONS						GAUGE BOSONS VECTOR BOSONS			
						SCALAR BOSONS			